Bike-Sharing Data Analysis Using Power BI

1. Data Import

- Imported the bike-sharing dataset into Power BI.
- Opened the dataset in Power Query for data cleaning.

2. Data Cleaning

- Sorted the number column in ascending or descending order as required.
- Removed the Position column because it was not required for the analysis.
- Checked for duplicate records, empty cells, and blank values.
- Cleaned and prepared the data for analysis.
- Loaded the cleaned data into Power BI.

3. DAX Measures

Created DAX measures to calculate the following:

- Total Number of Stations: Count of bike-sharing stations.
- Total Bike Stands: Total capacity of all bike stands.
- Total Available Bike Stands: Number of bike stands currently available for bikes.
- Total Available Bikes: Number of bikes currently available for users.

4. Dashboard Analysis

Created visualizations to analyze bike-sharing data effectively.

- Available Bikes by Contact Name: Shows bike availability for each contact or station manager.

- Bike Stands by Station Name: Compares the bike stand capacity of different stations.

- Monthly Bike Trend: Shows changes in bike availability over time, using the date information.

- Banking Availability: Shows the availability of stations with banking facilities.

5. Interactive Dashboard

- Added a Contact Name slicer to filter the dashboard by contact name.

- Added an Address slicer to filter the dashboard by location.

- These slicers help users explore the data and compare different stations.

6. Project Outcome

This project helped me understand how to import data; clean data using Power Query, create DAX measures, build charts, and develop an interactive dashboard in Power BI. The dashboard provides a clear view of bike availability, station capacity, monthly trends, and banking facilities.